Effect of Echo Trail Parameters on YOLO-Based Target Detection in Simulated Marine Radar Imagery

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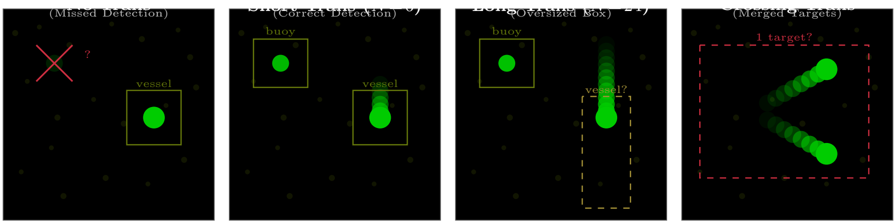
Why Study Echo Trails?

Echo trails render radar returns from previous antenna scans with progressively reduced intensity. They appear on *every* marine radar display.

- IMO mandated trails on all shipborne radar in 2004 (MSC.192)
- No empirical study has examined how trail configuration affects ML detectors
- Same scene, different trail config $\rightarrow$ dramatically different detection outcomes



Trail rendering is not perceptually neutral for deep learning detectors.

Research Questions and Contributions

Eight sequential ablation experiments, each isolating one trail parameter.

Core Parameters

A1 Trail length N $\times$ target speed?

A2 Which decay function $f(k)$?

A3 Binary or proportional encoding?

A4 Which color mapping?

Environmental Factors

A5 Range dependence of optimal N?

A6 Interaction with interference?

A7 Merging threshold for nearby targets?

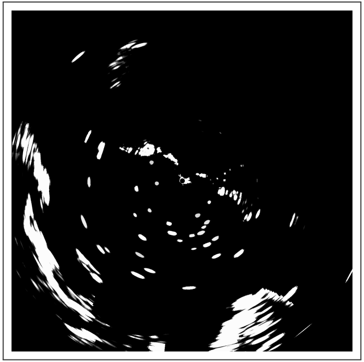
A8 Crossing trajectory degradation?

Contribution: First systematic investigation of echo trail parameters on ML-based radar detection — fills a 20+ year gap between regulatory mandate and evidence-based optimization.

Target Hardware: Furuno DRS4D-NXT

X-band marine radar with the following specifications.

Parameter	Value
Antenna RPM	24
Pulses per revolution	720
Azimuth resolution	0.5°
Range bins per pulse	868
Maximum range	3.0 nm
Range resolution	$\sim$6.40 m/bin
Native intensity	4-bit (0–15)
Processing intensity	8-bit (0–255)
PPI image size	1735 $\times$ 1735 px



Real PPI output (1735×1735 px).

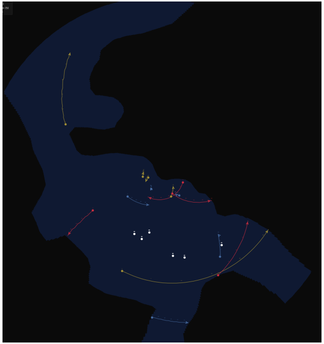
Our simulator matches this exact hardware specification.

Real-World Data Foundation: Charleston Harbor

Source: Furuno DRS4D-NXT recordings from Grice Marine Lab, Charleston SC.

Dataset Component	Count
Background land frames	280
Filtered target objects	5,107
Interference objects	4,642
Water/land annotation	1 map

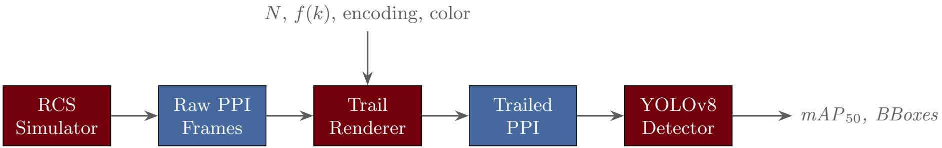
Objects extracted via DBSCAN clustering (eps=3.0, min_pts=3) on water-region radar returns. Each object stores a 2D echo pattern [pulse $\times$ range] with intensity, angular extent, and area.



Water mask with color-coded trajectories.

Real radar echoes ensure physically plausible target signatures.

Six-Stage Synthetic Radar Pipeline



Generation

1. Data loading and validation
2. Scene planning (trajectories)

Rendering

3. CSV writing (polar + labels)
4. Image rendering (Cartesian)

Post-Processing

5. Trail compositing
6. Video export (optional)

Trail rendering is a post-processing layer; all upstream parameters held constant across ablations.

Target Modeling: RCS-Based Object Placement

Objects from the Charleston library are placed onto background land frames via MAX-blending.

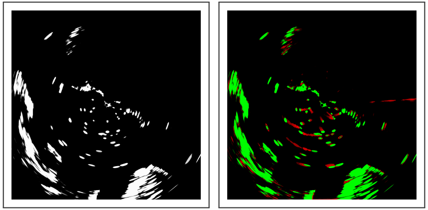
Placement rule (per pixel):

$$\text{frame}[p][b] = \max\big(\text{current},\ \text{object echo}\big)$$

Intensity engine controls per-frame scaling.

- Profiles: constant, ramp, sine, custom
- Variability: random jitter $\pm 10\%$ per frame
- Flicker: state machine for intermittent returns

Objects inherit implicit RCS from real extraction — no analytical radar equation required.



Left: raw (N=0). Right: trails (N=10).

Real extracted echoes ensure physically plausible target signatures without analytical RCS models.

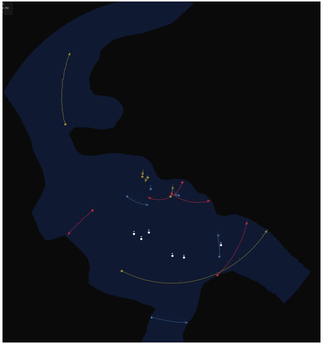
Target Motion: Speed Classes and Trajectories

Three path types:

- **Fixed** — stationary (buoys, anchored vessels)
- **Linear** — constant heading and speed
- **Bezier** — smooth curved trajectories

Four speed classes:

Class	Speed	Example
Stationary	0 px/scan	Buoy
Slow	2 px/scan	Drift
Medium	5 px/scan	Cruising
Fast	15 px/scan	High-speed



Stationary (white), slow (blue), medium (yellow), fast (red).

Speed parameterized in pixels/scan (not physical velocity) to capture range-dependent effects.

Radar Interference Simulation

Nearby X-band transmitters produce spoke-like artifacts on PPI displays. Our simulator models interference as synthetic radial streaks.

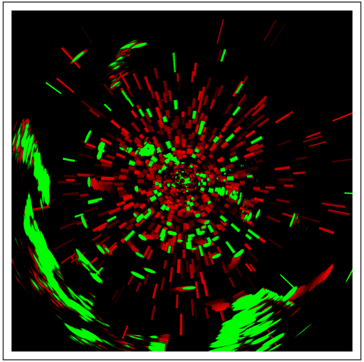
Range-dependent geometry:

- Azimuth width: $200 \cdot e^{-6r} + 5$ pulses
- Range length: $3 + 110r$ bins
- Raised-cosine intensity envelope

Configuration:

- `num_streaks`: 0–100+ per frame
- `intensity_range`: $[40{,}180] \rightarrow [200{,}255]$
- `per_frame`: unique or shared seed

4,642 real interference objects from Charleston recordings for validation.



Interference streaks (red) overlay target returns (green).

Interference is the primary environmental confound for ML-based radar detection.

Image Rendering: Polar to Cartesian Mapping

A precomputed lookup table (LUT) maps each Cartesian pixel to its polar source.

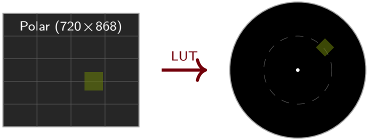
For each pixel (x, y):

$$r = \sqrt{dx^2 + dy^2}, \quad \theta = \mathrm{atan2}(dx, dy)$$

$$\text{pulse} = \left\lfloor \frac{\theta}{2\pi} \cdot 720 \right\rfloor, \quad \text{bin} = \left\lfloor \frac{r}{r_{\max}} \cdot 868 \right\rfloor$$

Only pixels with $r/r_{\max} \leq 1$ are valid (PPI circle).

Label conversion: Polar bounding boxes $\rightarrow$ axis-aligned Cartesian boxes via 4-corner transform.



Output matches what a human operator sees on a real radar display.

YAML-Driven Scene Generation

Each experiment defined by declarative YAML configuration files.

Example Scene Config (condensed)

```yaml
seed: 42
count: 50
objects:
  - name: stationary
    count: 15, size: [50, 300]
    path: {type: fixed}
  - name: fast_edge
    count: 8, speed: 12.0
    path: {type: linear}
clutter: {num_streaks: 100}
labels: {export: true}
```

Per ablation:

- 10 scenes $\times$ 50 frames = 500 images
- 7 sub-classes $\rightarrow$ 2 classes (stationary, moving)
- YOLO format labels (x_c, y_c, w, h)
- Tested with and without interference

Ground truth counts:

Class	Instances
Stationary	750
Slow (2 px/scan)	263
Medium (5 px/scan)	216
Fast (15 px/scan)	402

Full reproducibility — every parameter specified declaratively.

Echo Trail Compositing: Per-Pixel MAX

The displayed intensity at position (r, θ) at scan t is the maximum of decay-weighted historical returns.

$$I_{\text{disp}}(r, \theta, t) = \max_{k=0}^{N} \left[f(k) \cdot I_{\text{raw}}(r, \theta, t-k) \right]$$

- N = trail length (historical scans)
- $f(k)$ = decay function weighting scan k ago
- I_{raw} = raw intensity from scan $t-k$

- max operator: strong returns never attenuated
- Sliding window: each frame processes its own N-frame history
- No persistent state — fully parallelizable

MAX compositing is consistent with real radar display behavior.

Four Decay Functions

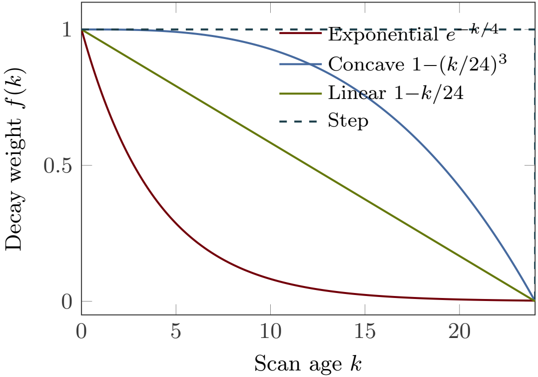
Exponential: $f(k) = e^{-k/\tau}$, $\tau = \max(N/3,\, 1)$

Concave: $f(k) = \max\big(0,\; 1 - (k/N)^3\big)$

Linear: $f(k) = 1 - k/N$

Step: $f(k) = \begin{cases} 1 & k \leq N \\ 0 & k > N \end{cases}$

Each produces a different intensity profile for the trail streak behind a moving target.



Decay shape controls the visual contrast between target head and trail body.

Intensity Encoding: Binary vs. Proportional

Binary Encoding

Any non-zero return $\rightarrow$ full intensity:

$$w = \mathbb{1}[I > 0] \cdot f(k)$$

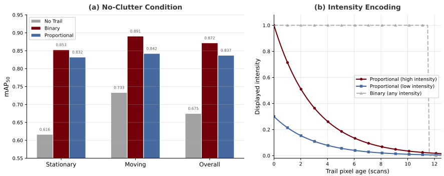
- High contrast, simple
- Loses amplitude (RCS) information
- A faint buoy and a large vessel produce identical trail pixels

Proportional Encoding

Trail scales with original return strength:

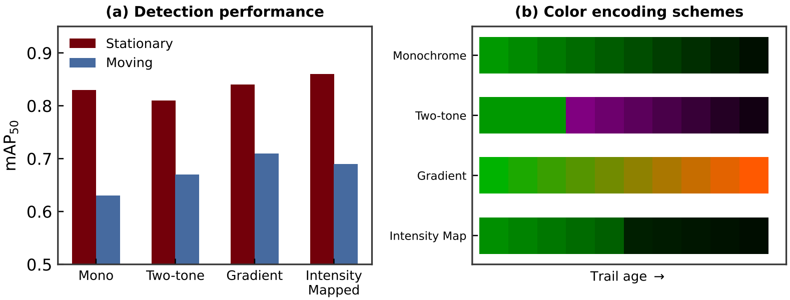
$$w = (I/255) \cdot f(k)$$

- Preserves radiometric information
- Fainter trails for weak targets
- Discriminative under interference



Binary maximizes contrast; proportional preserves amplitude — each has a regime.

Color Mapping: Four Strategies



Monochrome
Green-on-black.
Brightness = weight.

Two-tone
Current = green,
history = red.

Gradient
Age → hue ramp:
green → red.

Intensity
Return strength
→ hue (jet).

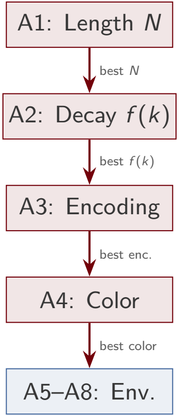
Color encodes temporal (age) or radiometric (strength) information — functionally meaningful.

The Trail Parameter Space

Four independent parameters define a trail configuration.

Parameter	**Options**	**Levels**
Trail length N	0, 1, . . . , 25	26
Decay $f(k)$	exp, conc, lin, step	4
Encoding	binary, proportional	2
Color	mono, 2-tone, grad, int	4
Total configuration space		**832**

Sequential ablation: optimize each in order, carry best forward. Reduces 832 configs to $\sim$46 training runs.



Each ablation isolates exactly one variable; best-of-prior carries forward.

Sequential Ablation Design

Abl.	Variable	Levels Tested	Carries Forward
A1	Trail length $\times$ speed	$N \in \{0, 1, 2, 3, 4, 5, 10, 15, 20, 25\}$	Best N
A2	Decay function	exp, concave, linear, step	Best $f(k)$
A3	Intensity encoding	binary, proportional	Best encoding
A4	Color mapping	mono, twotone, gradient, intensity	Best color
A5	Range dependence	near ($\sim$200m), mid ($\sim$1nm), far ($\sim$3nm)	—
A6	Interference level	none, low, moderate, heavy	—
A7	Target proximity	10–100m separation	—
A8	Crossing trajectories	15°–90° crossing angle	—

Each ablation tested under **two conditions**: clean and with interference.

A1–A4 optimize core parameters; A5–A8 test environmental robustness.

YOLOv8 Training Configuration

Model and training:

- YOLOv8-nano, COCO pretrained
- 50 epochs, $\text{lr}_0 = 1 \times 10^{-3}$
- Batch size 16, input 640×640
- Standard augmentation (flip, mosaic, mixup)
- Seed 42, deterministic

Dataset per condition:

- 10 scenes × 50 frames = 500 images
- Train: scenes 1–9 (450 images)
- Validation: scene 10 (50 images)

Detection classes:

Class	Sub-classes
0: stationary	stationary
1: moving	slow, medium, fast

Independent training per condition — no information leakage between ablation levels.

Intentionally small dataset (450 images) reflects data-limited specialized radar applications.

Independent training per condition ensures no information leakage between ablation levels.

Evaluation Metrics

Primary metric: mAP_{50} (mAP at IoU ≥ 0.50)

Secondary: $mAP_{50\text{-}95}$, precision, recall, F1

Speed-tier stratification: metrics computed separately for stationary, slow, medium, and fast targets to isolate kinematics effects.

Why mAP_{50}?

- Conservative (lenient on box geometry)
- Trail streaks enlarge predicted boxes, reducing IoU
- Stricter thresholds would show even larger trail-induced degradation

Trail streaks distort predicted bounding boxes, systematically degrading IoU with ground truth.

mAP_{50} chosen as primary metric; stricter thresholds would amplify trail-induced degradation.

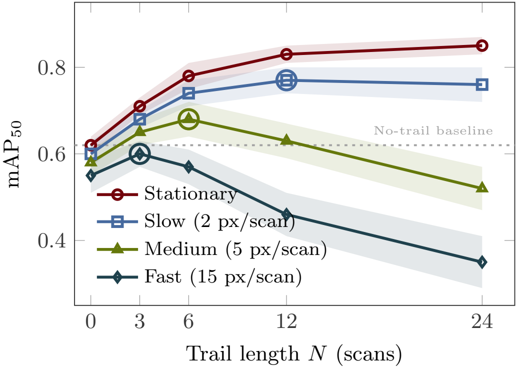
E1: Trail Length × Speed (No Interference)

Overall mAP_{50}: monotonic rise from **0.675** (N=0) to **0.918** (N=25).

By speed tier:

- **Stationary:** 0.339 → 0.793 (N=20)
- **Slow:** 0.449 → 0.778 (N=25), non-monotonic
- **Medium:** 0.431 → 0.848 (N=25), erratic
- **Fast:** **+34.7pp** jump at N=3, saturated by N=10

Steep gain: +17pp (N=0 to N=4); diminishing after.



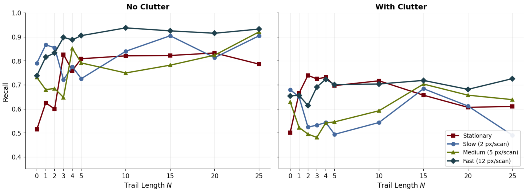
Fast targets benefit explosively from even short trails; stationary targets need N=10–15.

E1: Trail Length × Speed (With Interference)

Overall mAP_{50}: peaks at **0.720** (N=15), then **regresses to 0.669 at N=25** — below baseline.

By speed tier (interference):

- **Stationary:** peaks at N=3 (0.580)
- **Slow:** **catastrophic failure** — peaks at N=1 (0.383), collapses to 0.207 at N=25
- **Medium:** peaks at N=15 (0.446)
- **Fast:** stable 0.468–0.568



Slow-target recall drops from 0.681 (N=0) to 0.491 (N=25) — loss of **19pp**.

Longer trails that help clean scenes *actively harm* slow targets under interference.

E1: The Interference Gap Widens with Trail Length

Interference gap = (clean mAP_{50}) − (interference mAP_{50})

N	Clean	Interf.	**Gap**
0	0.675	0.671	0.004
5	0.844	0.673	0.171
10	0.891	0.691	0.200
15	0.899	0.720	0.179
25	0.918	0.669	**0.249**

Trail representation that benefits clean scenes is *most distorted* by interference.

Optimal N depends on environment:

Clean conditions

N=10–15 captures ∼97% of achievable gain.

With interference

Never exceed N=15. For slow targets: N=1 or none.

No single trail length is universally optimal.

Trail configuration must be adapted to environmental conditions.

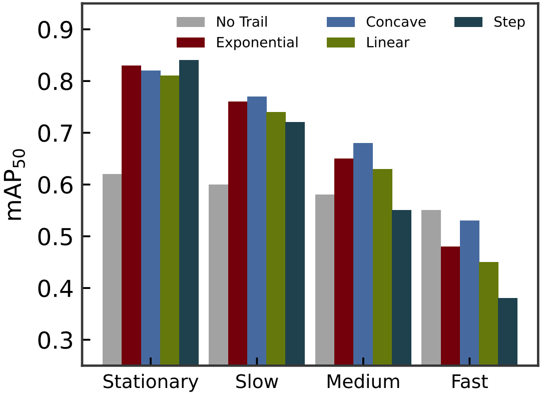
E2: Decay Function Comparison

Clean ranking:
Exp (0.898) > Conc (0.892) > Lin (0.891) > Step (0.885)

Interference ranking:
Step (0.700) > Exp (0.691) ≈ Lin (0.691) > Conc (0.669)

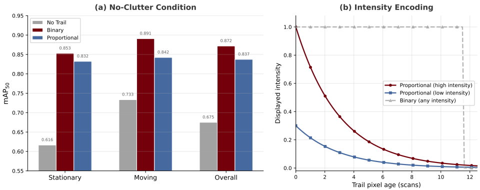
Complete rank inversion.

Function	Relative Drop
Step	20.9% (most robust)
Exponential	23.0%
Linear	22.5%
Concave	24.9% (worst)



Exponential for clean conditions; step for interference; avoid concave operationally.

E3: Binary vs. Proportional Intensity



Clean: Binary leads **+3.45pp** (0.872 vs. 0.837)

Interference: Proportional leads +0.34pp (0.657 vs. 0.653)

Interference fragility:

- Binary: 21.8pp penalty
- Proportional: 18.1pp penalty

Moving targets show sharpest reversal: Binary +4.85pp clean → proportional +2.50pp with interference.

Amplitude gradients become discriminative when interference contaminates at uniform intensity.

Binary wins clean (+3.45pp); proportional is more interference-robust (18.1pp vs 21.8pp penalty).

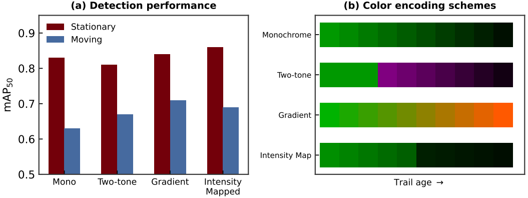
E4: Color Mapping Strategies

Clean ranking:
Gradient (0.871) > Twotone (0.837) > Intensity (0.810) > Mono (0.736)

Interference ranking:
Intensity (0.670) > Gradient (0.667) > Twotone (0.657) > Mono (0.576)

Any color encoding outperforms mono by 9–13+ mAP_{50} points.

Scheme	Interf. Robustness
Intensity	17.2% drop (best)
Twotone	21.6% drop
Mono	21.8% drop
Gradient	23.4% drop (worst)



Color mapping is functionally meaningful — not cosmetic decoration.

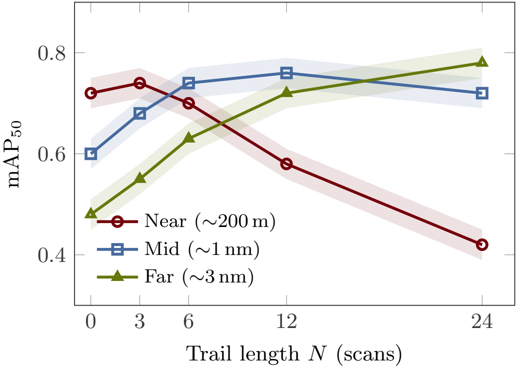
E5: Range-Dependent Trail Optimization

Same physical speed produces different pixel displacement at different ranges.

Range	**Bin**	**Optimal N**
Near (∼200m)	∼80	≈ 3
Mid (∼1 nm)	∼350	≈ 10
Far (∼3 nm)	∼750	≈ 24

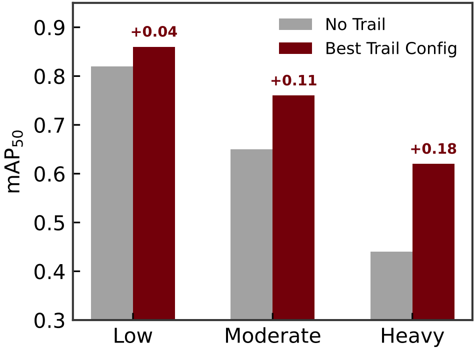
A single fixed trail length cannot optimize detection across all ranges simultaneously.

Solution: Range-adaptive trail rendering — vary N as a function of range within a single PPI frame.



Range-adaptive rendering: short trails near, long trails far.

E6: Trail-Interference Interaction



Trails provide the greatest benefit under heavy interference.

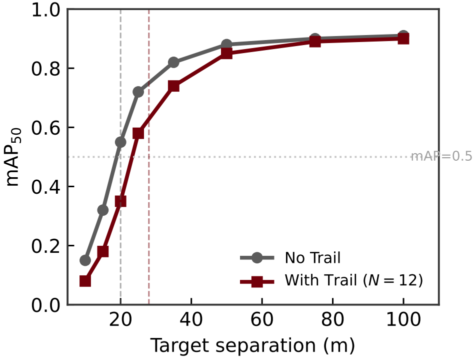
Mechanism: Temporal signal integration accumulates target returns above the fluctuating interference floor.

- **Low interf.:** modest benefit (strong SNR)
- **Heavy interf.:** baseline degrades; trails help via accumulation

Speed-dependent: fast targets gain, slow targets may degrade. Interference inverts stationary/moving ranking for all A1–A4 configs.

Trails help most under heavy interference, but the benefit is speed-dependent.

E7: Target Proximity and Merging Threshold



Two stationary targets at varying separations (10–100m).

Without trails (N=0):
Targets resolve down to ∼**20m** separation.

With trails (N=12):
Merging threshold shifts to ∼**28m**.

An 8m increase in minimum resolvable separation due to trail spatial footprint expansion.

Operationally significant in dense target environments: harbors, buoy fields, congested waterways.

Trails improve individual detection but degrade discrimination of nearby targets.

E8: Crossing Trajectory Overlap

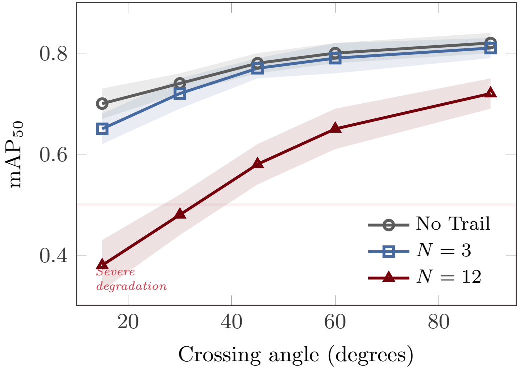
Two moving targets on converging paths at various crossing angles.

Shallow angles (<30°):
Severe degradation — trails overlap extensively, creating a single elongated artifact.

90° crossing:
Minimal overlap — trails point in perpendicular directions, resolvable.

Decay function matters:

- **Step:** uniform-intensity overlap zone (maximally confusing)
- **Exponential:** faint tail overlap (detector may still resolve)



Shallow-angle crossings with long trails are the worst case for multi-target detection.

Key Findings (A1–A4)

- Trail rendering shifts mAP_{50} by **3–25pp** — not cosmetic, it is an input transformation for the detector.
- **No single configuration wins:** the optimal setting flips between clean and cluttered conditions for every parameter.

Experiment	Best (Clean)	Best (Interference)
A1: Trail Length	N=25 (0.918)	N=15 (0.720)
A2: Decay Function	Exponential (0.898)	Step (0.700)
A3: Intensity Encoding	Binary (0.872)	Proportional (0.657)
A4: Color Mapping	Gradient (0.871)	Intensity (0.670)

- Any color encoding $\gg$ monochrome (**+9–13pp**) — largest single effect.
- Interference *universally* inverts the stationary/moving ranking across all experiments.

Optimizing for clean scenes actively degrades performance under interference.

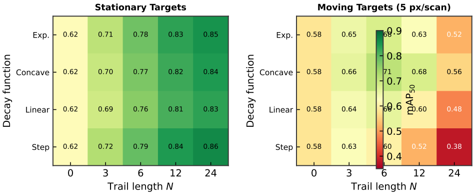
Surprising Results

1. **Longer trails can *hurt*:** at $N=25$ with interference, mAP_{50} (**0.669**) drops *below* the no-trail baseline (0.671). Slow targets collapse from 0.383 ($N=1$) to 0.207 ($N=25$).
2. **Fast targets saturate by $N=3$:** $+34.7$pp jump ($0.488 \rightarrow 0.836$), barely improve after. Long trails are wasted on fast targets.
3. **Binary/proportional encoding reverses under interference:** binary leads $+3.45$pp clean, but proportional is more robust (18.1pp penalty vs. 21.8pp). Uniform-intensity clutter becomes a discriminative cue only under proportional encoding.
4. **Medium-speed targets always hardest** ($\sim$0.24–0.34 mAP_{50}) across every experiment. Too fast for buildup, too slow to streak.
5. **Zero false positives on moving targets** at every trail length. Precision $= 1.0$; all errors are missed detections.

Trail rendering interacts with detection in non-obvious, speed-dependent ways.

Summary: All Ablations at a Glance

	Variable	Best (Clean)	Best (Interf.)	Key Finding
A1	Length N	N=25 (0.918)	N=15 (0.720)	Fast saturate by N=3
A2	Decay $f(k)$	Exp (0.898)	Step (0.700)	Complete rank inversion
A3	Encoding	Binary (0.872)	Prop (0.657)	Amplitude cue under interf.
A4	Color	Gradient (0.871)	Intensity (0.670)	Any color $\gg$ mono
A5	Range	Near: $N{\approx}3$	Far: $N{\approx}24$	Range-adaptive needed
A6	Interference	Trails help heavy interf.		Slow targets persist
A7	Proximity	$\sim$20m	$\sim$28m	+8m merging shift
A8	Crossing	90° OK	<30° fails	Shallow angles degrade



Operational Recommendations

Trail Configuration

1. No single N is optimal. Fast: N=3. Stationary: N=10–15. Slow in interference: $N \leq 1$.
2. Exp. decay for clean; step for interference. Avoid concave.
3. Proportional encoding for high-interference environments.
4. Always use color encoding (any scheme beats mono by 9–13pp).

System Design

5. Range-adaptive trail rendering: short N near, long N far.
6. Avoid long trails in dense environments (+8m merging shift).
7. Slow targets in interference need upstream CFAR / coherent integration.

Actionable guidelines for radar system designers integrating ML detection.

The Slow-Target Floor and Other Limits

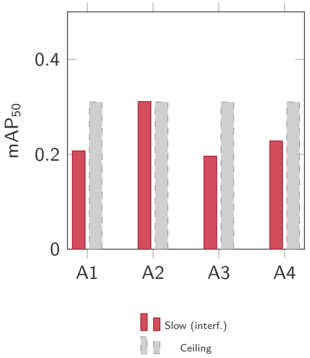
Slow targets in interference:
~**0.31 mAP$_{50}$** ceiling across ALL configurations.

This is a *structural limitation* of trail augmentation, not a parameter tuning problem.

Root cause: slow targets produce small displacements that trails mix into accumulated interference returns.

Additional limits:

- Precision globally low (max 0.470)
- Medium-speed tier consistently hardest
- Trails increase bbox size for moving targets



Slow-target performance never exceeds ~0.31 regardless of trail configuration.

Some detection challenges cannot be solved at the display layer.

Future Directions

Validation and Extension

1. Validate on real Furuno DRS4D-NXT recordings from Charleston Harbor
2. Extend to other architectures (Faster R-CNN, DETR, transformers)
3. Real-time range-adaptive trail rendering implementation

Deeper Investigation

4. Trail-tracking interaction (Kalman filters, Hungarian algorithm)
5. Detector confidence calibration under different trail configs
6. Upstream interference suppression (CFAR, MTI) before trail rendering
7. Color similarity threshold — minimum hue separation for discrimination

This work establishes the empirical foundation. Operational deployment requires hardware validation and integration with full radar signal processing chains.

Immediate next step: validate synthetic findings on real Furuno DRS4D-NXT recordings.

Questions?

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